Online Resource 1

Supplementary tables and CLAIM checklist

Segmentation, type classification and measurement of brain tumors on single 2D MRI slices: a retrospective development and leakage-controlled internal validation study

Journal of Imaging Informatics in Medicine. Author details are given on the title page (double-blind review).

Table S1 Completed CLAIM checklist

Item	Topic	Where reported
1	Identification as a study of AI methodology	Title; Abstract
2	Structured summary	Abstract
3	Scientific and clinical background, intended use	Introduction
4	Study objectives and hypotheses	Introduction, last paragraph
5	Prospective or retrospective study	Methods: Study design (retrospective)
6	Study goal (model creation, exploratory, feasibility, non-inferiority)	Methods: Study design (development and internal validation)
7	Data sources	Methods: Datasets
8	Eligibility criteria	Methods: Datasets (all slices of both public datasets used)
9	Data pre-processing	Methods: Preprocessing
10	Selection of data subsets	Methods: Data splits and leakage control; Table S2
11	Definitions of data elements	Methods: Datasets; Online Resources 2-4 (column definitions)
12	De-identification methods	Title page, Statements and Declarations: Ethics (public de-identified data)
13	How missing data were handled	Not applicable: no missing images or masks in figshare; Kaggle has no masks by design (Methods: Training)
14	Definition of ground truth reference standard	Methods: Datasets (expert masks; dataset class labels)
15	Rationale for choosing the reference standard	Methods: Datasets; Discussion (expert masks and their extent)
16	Source of ground-truth annotations; qualifications of annotators	Methods: Datasets (annotations by the dataset creators [6-8]; qualifications as reported there)
17	Annotation tools	Not applicable: masks taken from the public dataset
18	Measurement of inter- and intrarater variability	Not available for the public masks; stated as limitation of reference standards (Introduction [4])
19	Intended sample size and how it was determined	Methods: Data splits (all available data; no sample-size calculation)
20	How data were assigned to partitions; proportions	Methods: Data splits and leakage control; Online Resource 2
21	Level at which partitions are disjoint	Methods: Data splits (patient level for figshare; image-level copy removal for Kaggle)
22	Detailed description of model	Methods: Model
23	Software libraries, frameworks and packages	Methods: Training; Table S3; title page (Code availability)
24	Initialization of model parameters	Methods: Model (ImageNet weights for the encoder)
25	Details of training approach (augmentation, hyper-parameters, stopping)	Methods: Training; Table S3
26	Method of selecting the final model	Methods: Training (best validation score)
27	Ensembling techniques	Methods: Inference (horizontal-flip averaging only); Table S6
28	Metrics of model performance	Methods: Evaluation and statistics
29	Statistical measures of significance and uncertainty	Methods: Evaluation and statistics (bootstrap 95% CI)
30	Robustness or sensitivity analysis	Results: Calibration, ablations; Table S6
31	Methods for explainability or interpretability	Results: Fig. 2 and Fig. 6 (outlines shown with the prediction)
32	Validation or testing on external data	Not performed; stated in Limitations
33	Flow of participants or cases	Methods: Data splits; Table S2
34	Demographic and clinical characteristics of cases	Not available in the public datasets; tumor-type composition in Methods: Datasets
35	Performance metrics for optimal model on all data partitions	Results; Tables 1-3; Fig. 5 (validation curves)
36	Estimates of diagnostic accuracy and their precision	Results; Table 3 (95% CI)

Item	Topic	Where reported
37	Failure analysis of incorrectly classified cases	Results: Fig. 2 (worst cases); Discussion; Online Resources 3-4
38	Study limitations	Discussion: Limitations
39	Implications for practice	Discussion; Conclusions
40	Registration number and name of registry	Not applicable (retrospective study on public data)
41	Where the full study protocol can be accessed	Methods; source code repository (title page)
42	Sources of funding and other support	Title page, Statements and Declarations: Funding

CLAIM, Checklist for Artificial Intelligence in Medical Imaging (Mongan et al., Radiol Artif Intell 2020). Section names refer to the main manuscript; numbers in brackets are its references. Item wording is abbreviated; the official checklist should be consulted for the full text.

Table S2 Dataset partitions and leakage control

Partition	Size	Use
figshare training	186 patients / 2439 slices	Segmentation and type losses
figshare validation	23 patients / 330 slices	Model selection
figshare test	24 patients / 295 slices	Segmentation, measurement, side and type evaluation
Kaggle Training, removed	305 images	Copies of figshare validation or test slices
Kaggle training subset	4867 images	Classification loss (no-tumor images also segmentation loss)
Kaggle validation subset	540 images	Model selection
Kaggle Testing, all	1311 images	Classification and detection evaluation
Kaggle Testing, copies of a training image	621 images	Reported separately
Kaggle Testing, no copies	690 images (no tumor 28, glioma 243, meningioma 177, pituitary 242)	Leakage-controlled subset

Copies were identified by comparing 32 × 32 z-scored thumbnails (mean absolute difference < 0.08). The patient-level split is listed in Online Resource 2.

Table S3 Training set-up and software

Item	Setting
Input	256 × 256 grayscale, zero-padded to square, scaled to [0, 1]
Architecture	U-Net decoder, ResNet34 encoder (ImageNet weights), 24.4 million parameters
Heads	Segmentation (1 channel, sigmoid); classification (4 classes, dropout 0.3)
Loss	BCE + soft Dice (segmentation) + 0.5 × weighted cross-entropy, label smoothing 0.05; class weights 1, 1, 1.2, 1
Optimizer / schedule	AdamW, weight decay 1e-4; one-cycle, maximum learning rate 3e-4, 10% warm-up
Batch size / epochs	12 / up to 12 (selected epoch 11)
Data per epoch	All figshare training slices + 1500 random Kaggle training images
Augmentation	Horizontal flip p = 0.5; rotation ±15°; scale 0.85–1.15; translation ±8%; gamma 0.7–1.5; contrast ×0.85–1.15; brightness ±0.08; Gaussian blur p = 0.2; Gaussian noise p = 0.3
Selection criterion	Validation Dice (figshare) + validation accuracy (Kaggle)
Inference	Average with horizontally flipped input; mask threshold 0.5
Hardware	12-core CPU, no GPU; about 13 min per epoch
Software	Python 3.12.10; PyTorch 2.5.1; torchvision 0.20.1; NumPy 2.3.5; SciPy 1.17.0; scikit-image 0.26.0; OpenCV 4.13.0; scikit-learn 1.8.0
Random seed	Patient split 42; training 0

BCE, binary cross-entropy; CPU, central processing unit.

Table S4 Per-class classification metrics on the Kaggle Testing folder

Class	Images	Precision	Recall	F1
No tumor	405	0.998	0.995	0.996
Glioma	300	0.987	0.987	0.987
Meningioma	306	0.990	0.977	0.984

Class	Images	Precision	Recall	F1
Pituitary tumor	300	0.980	0.997	0.988

Four-class accuracy 98.9% (95% CI 98.4–99.5%), macro-F1 0.989. Confusion matrix rows (true class: predicted no tumor / glioma / meningioma / pituitary) — No tumor: 403 / 1 / 1 / 0; Glioma: 0 / 296 / 1 / 3; Meningioma: 1 / 3 / 299 / 3; Pituitary tumor: 0 / 0 / 1 / 299.

Table S5 Measurement, side and scale-estimation details

Quantity	Result
Longest diameter, all outlines (n = 292)	median absolute error 3.6 mm; mean 9.4 mm; bias -0.8 mm; 95% limits -32.2 to 30.6 mm; r = 0.72
Longest diameter, overlapping outlines (n = 269)	median absolute error 3.0 mm; mean 8.6 mm; bias -0.1 mm; 95% limits -30.1 to 29.9 mm; r = 0.75
Tumor area	r = 0.79; median relative error 14.6%
Side agreement	90.4% of 104 eligible slices; 100% of 90 slices with overlapping outline
Scale estimate / nominal spacing	ratio 10th, 50th, 90th percentile 0.73, 0.84, 0.98
Low-confidence flag (mean probability < 0.8)	39 of 618 validation and test outlines missed the tumor; flagged 15 of these and 11 correct outlines
Segmentation failures	26 slices with Dice < 0.1; 3 slices without outline
Run time	0.15 s median (90th percentile 0.17 s) per image on 6 CPU threads

Diameters were measured on predicted and expert masks with the nominal 0.98 mm pixel size of the 256-pixel figshare images. Limits are Bland–Altman 95% limits of agreement.

Table S6 Effect of mask threshold and test-time flipping on Dice

Threshold	Dice mean, no flipping	Dice median, no flipping	Dice mean, flipping	Dice median, flipping
0.3	0.756	0.870	0.755	0.860
0.5	0.755	0.866	0.750	0.863
0.7	0.753	0.863	0.743	0.865

figshare test patients (24 patients, 295 slices). The manuscript reports the setting with flipping and threshold 0.5.